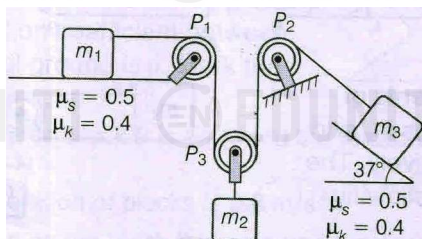


Passage (Q 14 and 15)

The situation shown in the figure, pulleys P_1 and P_2 are fixed while smooth and light pulley P_3 is movable. Three blocks m_1 , m_2 and m_3 of masses 10 kg, 8 kg and 2 kg respectively, are connected through a light string. ($g = 10 \text{ m/s}^2$)



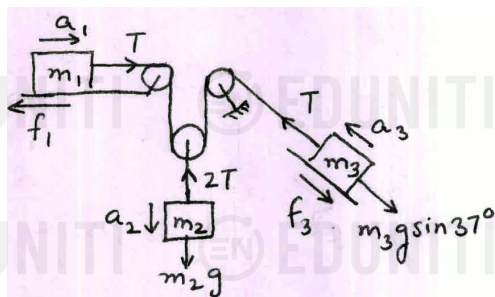
14. With reference to the information given in the passage, mark the correct option.
- m_1 remains stationary but m_3 is moving up along the inclined plane
 - m_2 is moving upward
 - All blocks remain stationary
 - m_1 moves rightward
15. The acceleration of block m_2 is
- 2 m/s^2
 - 3 m/s^2
 - 2.7 m/s^2
 - 4.8 m/s^2

Passage (Solution)

- 14.(a) From FBD, $2T = m_3(g - a_2)$

T is maximum when $a_2 = 0$

$$T_{\max} = m_3 g / 2 = 8 \times 10 / 2 = 40 \text{ N}$$



For m_1 to move, $T_{\min} = \mu_s m_1 g = 0.5 \times 10 \times 10 = 50 \text{ N}$.

As $T_{\max} < 50 \text{ N}$, m_1 remains stationary.

For m_3 to move,

$$T_{\min} = m_3 g (\sin 37^\circ + \mu_s \cos 37^\circ) \\ = 2 \times 10 (0.6 + 0.5 \times 0.8) = 20 \text{ N}$$

As $T_{\max} > 20 \text{ N}$, m_3 moves up the inclined plane.

- 15.(c) $m_2 a_2 = m_2 g - 2T$ (1)

As m_3 is moving, the friction force on it is kinetic. Also,

$$a_3 = 2a_2$$

$$\Rightarrow m_3 a_3 = T - m_3 g (\sin 37^\circ + \mu_k \cos 37^\circ)$$

$$\Rightarrow m_3 (2a_2) = T - m_3 g (0.6 + 0.4 \times 0.8) \quad \text{.....(2)}$$

From (1) and (2),

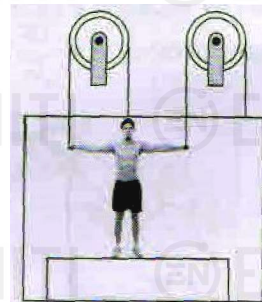
$$(m_2 + 4m_3) a_2 = (m_2 - 1.84 m_3) g$$

$$\Rightarrow (8 + 4 \times 2) a_2 = (8 - 1.84 \times 2) \times 10$$

$$\therefore a_2 = 2.7 \text{ m/s}^2$$

Passage (Q 16 and 17)

A weighing machine of mass 20 kg is fixed in a lift of mass 10 kg. A man of mass 60 kg is standing on the weighing machine and is pulling the lift through light strings passing over light fixed smooth pulleys. The force exerted on each string is same. ($g = 10 \text{ m/s}^2$)



16. If lift is stationary, then reading of weighing machine is
- 60 kg
 - 120 kg
 - 80 kg
 - 15 kg
17. If reading of weighing machine is equal to true weight of man, the force exerted on string by each hand of man is
- 1800 N
 - 900 N
 - 450 N
 - zero

Passage 1 (Solution)

- 16.(d) Mass of man, $m_M = 60 \text{ kg}$

Mass of weighing machine, $m_W = 20 \text{ kg}$

Mass of lift, $m_L = 10 \text{ kg}$

For the system,

$$4T = (m_M + m_W + m_L)g = (60 + 20 + 10) \times 10$$

$$\Rightarrow T = 225 \text{ N}.$$

For the man, $2T + N = m_M g = 600$

$$\Rightarrow N = 600 - 2T = 600 - 2 \times 225 = 150 \text{ N} = 15 \text{ kg wt}$$

$\therefore$ Reading = 15 kg

- 17.(b) Reading = True weight

$$\Rightarrow N = m_M g = 600 \text{ N}$$

Now the man is pulling the string with a larger force and the system is accelerating upwards with acceleration (say a).

For the man,

$$N + 2T - m_M g = m_M a \quad \Rightarrow 600 + 2T - 600 = 60a$$

$$\Rightarrow T = 30a$$

....(1)

For the system,

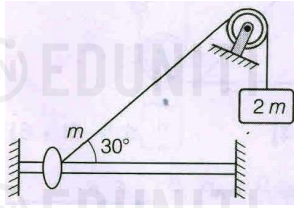
$$4T - 900 = 90a$$

....(2)

From (1) and (2), $T = 900 \text{ N}$

Passage (Q 18 and 20)

A smooth ring of mass $m = 1\text{ kg}$ can slide on a fixed horizontal rod. A light string tied to the ring passes over a fixed smooth pulley of mass $M = 0.2\text{ kg}$ and carries a block of mass $2m$ as shown in the figure. The ring is released from rest at the instant shown in figure. ($g = 10\text{ m/s}^2$)

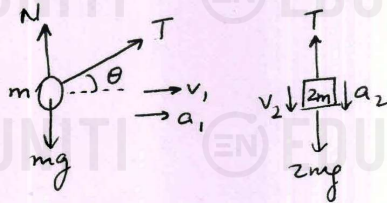


18. The tension in string at instant shown in the figure, is
(a) 8 N (b) 20 N
(c) 10 N (d) 15 N
19. The normal force on rod due to ring is
(a) 8 N (b) 6 N
(c) 10 N (d) 4 N
20. Reaction force (i.e. clamp force) on the pulley due to clamp is
(a) $8\sqrt{3}\text{ N}$ (b) 8 N
(c) $\sqrt{244}\text{ N}$ (d) none of these

Passage (Solution)

- 18.(a) The velocity along the string at the two ends remain the same.

$$\Rightarrow v_2 = v_1 \cos \theta$$



$$\Rightarrow \frac{dv_2}{dt} = \frac{dv_1}{dt} \cos \theta + v_1 \times \frac{d(\cos \theta)}{dt}$$

$$\Rightarrow a_2 = a_1 \cos \theta + 0 \times d(\cos \theta) / dt$$

$$\Rightarrow a_2 = a_1 \cos \theta$$

$$\text{From FBD, } ma_1 = T \cos \theta$$

$$\text{From (1), (2) and (3),}$$

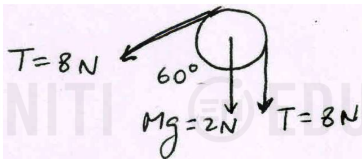
$$T = \frac{2mg}{1 + 2 \cos^2 \theta} = \frac{2 \times 1 \times 10}{1 + 2 \cos^2 30^\circ} = 8\text{ N}$$

- 19.(b) From FBD of ring,

$$N + T \sin \theta = mg$$

$$\therefore N = mg - T \sin \theta = 1 \times 10 - 8 \times \sin 30^\circ = 6\text{ N}$$

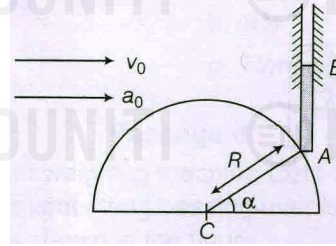
- 20.(c) Reaction force on pulley due to clamp



$$= \sqrt{(8+2)^2 + 8^2 + 2(8+2) \times 8 \times \cos 60^\circ} = \sqrt{244}\text{ N}$$

Passage (Q 21 and 22)

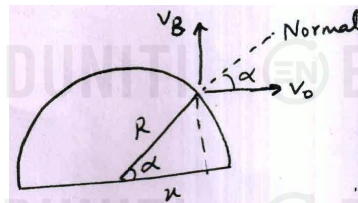
A smooth rod AB is subjected to vertical up and down movement system, while resting on a hemisphere of radius R . The hemisphere moves to the right with velocity v_0 and an acceleration of a_0 .



21. The velocity of the point B of the rod is
(a) $v_0 \tan \alpha$ (b) $v_0 \sin \alpha$
(c) $v_0 \cot \alpha$ (d) $v_0 \cos \alpha$
22. The acceleration of the point B of the rod is
(a) $a_0 \cot \alpha$ (b) $\frac{a_0 R \sin \alpha + v_0^2 + v_0^2 \tan^2 \alpha}{R \cos \alpha}$
(c) $\frac{a_0 R \cos \alpha + v_0^2 + \tan^2 \alpha}{R \cos \alpha}$ (d) $a_0 \cot \alpha - \frac{v_0^2}{R} \operatorname{cosec}^3 \alpha$

Passage (Solution)

- 21.(c) The velocity along the normal remains the same.



$$\Rightarrow v_0 \cos \alpha = v_B \sin \alpha \quad \therefore v_B = v_0 \cot \alpha$$

- 22.(d) $\frac{dv_B}{dt} = \frac{dv_0}{dt} \cot \alpha - v_0 \operatorname{cosec}^2 \alpha \frac{d\alpha}{dt}$ (1)

$$\text{Also, } x = R \cos \alpha$$

$$\Rightarrow -v_0 = \frac{dx}{dt} = -R \sin \alpha \frac{d\alpha}{dt} \quad \dots\dots(2)$$

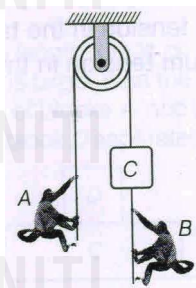
$$\text{From (1) and (2),}$$

$$a_B = a_0 \cot \alpha - v_0 \operatorname{cosec}^2 \alpha \times \frac{v_0}{R} \operatorname{cosec} \alpha$$

$$= a_0 \cot \alpha - \frac{v_0^2}{R} \operatorname{cosec}^3 \alpha$$

Passage (Q 23 and 24)

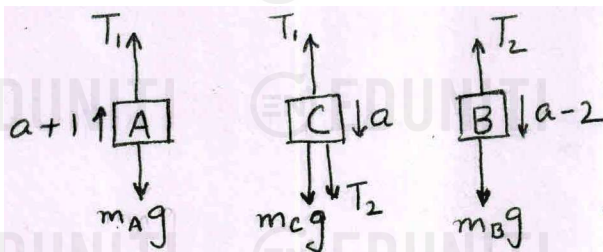
For the situation shown in figure, the monkey A of mass 20 kg climbs up the light rope with an acceleration 1 m/s^2 relative to the rope. The monkey B of mass 20 kg climbs up the light rope with an acceleration of 2 m/s^2 relative to rope. The mass of block C is 10 kg.



23. The acceleration of the block C is
 (a) 1 m/s^2 (b) 2 m/s^2
 (c) 2.4 m/s^2 (d) $\sqrt{5} \text{ m/s}^2$
24. Initially, the monkey A is 1.7 m below the fixed pulley. The system is released from rest. The time taken by the monkey A to reach the pulley is
 (a) 0.6 s (b) 1 s
 (c) 0.3 s (d) 2 s

Passage (Q17 to 18) Solution

- 23.(c) Let a be the upward acceleration of left side of the rope.



Then, acceleration of monkey A is $a + 1$ upwards, of monkey B is $a - 2$ downwards and of block C is a downwards.

From FBD,

$$T_1 - m_A g = m_A (a + 1) \Rightarrow T_1 - 200 = 20(a + 1) \dots (1)$$

$$m_B g - T_2 = m_B (a - 2) \Rightarrow 200 - T_2 = 20(a - 2) \dots (2)$$

$$m_C g + T_2 - T_1 = m_C a \Rightarrow 100 + T_2 - T_1 = 10a \dots (3)$$

From (1), (2) and (3),

$$100 = 50a - 20 \Rightarrow a = 2.4 \text{ m/s}^2$$

24.(b) For block A, $s = \frac{1}{2}(a + 1)t^2$

$$\therefore t = \sqrt{\frac{2s}{a+1}} = \sqrt{\frac{2 \times 1.7}{2.4+1}} = 1 \text{ s}$$